

Full Stack AI Engineer Roadmap (GenAI + Full Stack Development)

Full Stack Engineer + Cloud Basics + GenAI + RAG Roadmap (5 Months)

You Know MERN Stack → Industry-Ready in 5 Months



OVERALL TIMELINE (20 Weeks)

Month	Focus	Weekly Hours	What You Already Have
Month 1	Backend Excellence + PostgreSQL + Redis	25–30 hrs	MERN (React + Node + MongoDB)
Month 2	DSA Fundamentals + System Design	25–30 hrs	MERN basics
Month 3	GenAI Basics + LLM APIs + Prompt Engineering	25–30 hrs	Full Stack basics
Month 4	RAG + LangChain + Vector Databases	25–30 hrs	GenAI basics
Month 5	Cloud + Docker + CI/CD + Capstone Projects	30–35 hrs	Full Stack + GenAI + RAG

Daily Schedule: 4–5 hours (2 hrs Dev/DSA, 1.5 hrs GenAI/RAG, 1 hr Cloud)



MONTH 1: Backend Excellence + PostgreSQL + Redis (Weeks 1–4)

You know MERN, but need to go beyond MongoDB for production apps

Week 1–2: PostgreSQL Deep Dive

Topic	What to Learn	Practice
Basics	CREATE TABLE, INSERT, SELECT, UPDATE, DELETE	Create 5 tables
Constraints	PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK	Add constraints to 3 tables
Joins	INNER, LEFT, RIGHT, FULL, CROSS	Write 10 JOIN queries
Aggregations	GROUP BY, COUNT, SUM, AVG, MIN, MAX	Write 10 aggregation queries
Subqueries	IN, EXISTS, derived tables	Write 5 subquery solutions
Indexes	CREATE INDEX, WHERE indexes help	Index 3 tables
Transactions	BEGIN, COMMIT, ROLLBACK, ACID	Write 3 transaction examples
Prisma ORM	Schema, models, queries, migrations	Connect Prisma to 2 apps

YouTube Resources:

- **FreeCodeCamp**: "PostgreSQL Full Course" (5 hours)
- **CodeWithHarry**: "PostgreSQL Complete Course in Hindi" (4 hours)
- **Prisma**: "Prisma ORM Tutorial" (3 hours)
- **Kudening**: "PostgreSQL for Beginners" (3 hours)

Project 1: E-commerce Backend with PostgreSQL

- User registration/login (JWT + bcrypt)
 - Product CRUD (PostgreSQL + Prisma)
 - Shopping cart
 - Orders with transactions
 - 10+ REST API endpoints
 - Replace MongoDB with PostgreSQL
-

Week 3: Redis Caching

Topic	What to Learn	Practice
Basics	SET, GET, DEL, EXPIRE	Cache 5 data items
Data Types	String, Hash, List, Set	Use 3 types in app
Caching Strategies	Cache-aside, write-through	Cache 2 API endpoints
Pub/Sub	Publish, subscribe	Build 1 chat feature
Integration	Node.js + ioredis	Connect Redis to backend

YouTube Resources:

- **FreeCodeCamp:** "Redis Full Course" (2 hours)
- **Net Ninja:** "Redis Tutorial" (2 hours)
- **CodeWithHarry:** "Redis Complete Course in Hindi" (2 hours)

Project 1 Update: Add Redis to E-commerce

- Cache product listings
 - Cache user sessions
 - Add Pub/Sub for real-time notifications
-

Week 4: Authentication + Security + Middleware

Topic	What to Learn	Practice
JWT	Token generation, validation, expiration	Implement auth in 2 apps
bcrypt	Hash, compare, salt	Secure passwords in 2 apps
OAuth2	Google login, access tokens	Add Google auth to 1 app
Middleware	Express middleware chain	Create 3 custom middleware
Rate Limiting	express-rate-limit	Add to 2 APIs
Helmet	Security headers	Secure 2 APIs

CORS	Configure allowed origins	Fix CORS in 2 apps
-------------	---------------------------	--------------------

YouTube Resources:

- **FreeCodeCamp:** "JWT Authentication Full Course" (2 hours)
- **The Net Ninja:** "Authentication Tutorial" (3 hours)
- **Web Dev Simplified:** "JWT with Express" (2 hours)

Project 1 Final: Complete E-commerce with Security

- JWT authentication + bcrypt
- Google OAuth2 login
- Rate limiting middleware
- Helmet security headers
- CORS configured
- 15+ secure API endpoints

MONTH 2: DSA Fundamentals + System Design (Weeks 5–8)

Week 5–6: DSA Basics (C++ - Keep Your Current Path)

Topic	What to Learn	LeetCode Problems
Arrays	2-pointer, sliding window, prefix sum	Two Sum, Best Time to Buy Stock, Contains Duplicate
Strings	Palindrome, reversal, frequency	Valid Palindrome, Reverse Words, Valid Anagram
Hashing	HashMap, HashSet, frequency	Group Anagrams, Top K Frequent Elements
Linked List	Reverse, merge, cycle detection	Reverse LL, Merge Two Sorted Lists, Linked List Cycle

YouTube Resources:

- **FreeCodeCamp:** "DSA Full Course" (12 hours)
- **CodeWithHarry:** "DSA Complete Course in Hindi" (15 hours)
- **NeetCode:** "DSA Roadmap" (2 hours)
- **TakeUForward:** "DSA in 30 Days" (10 hours)

Target: 100 LeetCode (Easy: 60, Medium: 40)

Week 7: DSA Intermediate

Topic	What to Learn	LeetCode Problems
Stacks	Push, pop, parentheses, monotonic	Valid Parentheses, Min Stack, Daily Temperatures
Trees	DFS, BFS, traversal, BST	Maximum Depth, Invert Tree, Level Order, Validate BST
Recursion	Base case, recursive tree	Factorial, Fibonacci, Subsets, Permutations

Target: 200 LeetCode (Easy: 80, Medium: 100, Hard: 20)

Week 8: System Design Basics

Topic	What to Learn	Practice
REST APIs	GET, POST, PUT, DELETE, status codes	Design 3 APIs
Database Design	ER diagrams, normalization	Design 2 databases
Caching	Where to cache, TTL	Cache 2 apps
Load Balancing	Round-robin, sticky sessions	Explain to yourself
Microservices	Monolith vs microservices	Design 1 microservice

YouTube Resources:

- **FreeCodeCamp:** "System Design for Beginners" (4 hours)
- **Gaurav Sen:** "System Design Basics" (3 hours)
- **CodeWithHarry:** "System Design in Hindi" (4 hours)

Project 2: Social Media App (Full Stack)

- User profiles
- Posts with images
- Comments

- Follow/unfollow
 - Like system
 - PostgreSQL + Redis
 - JWT auth
 - 12+ API endpoints
-

MONTH 3: GenAI Basics + LLM APIs + Prompt Engineering (Weeks 9–12)

Week 9: Python Basics (2 Weeks - Only for AI)

Topic	What to Learn	Practice
Variables	int, float, string, list, dict	Write 10 programs
Functions	def, parameters, return, lambda	Create 10 functions
Loops	for, while, break, continue	Write 10 iteration programs
Classes	class, <code>__init__</code> , methods	Create 3 classes
Libraries	numpy, pandas, requests, json	Use 3 libraries

YouTube Resources:

- **FreeCodeCamp**: "Python for Beginners" (4 hours)
- **CodeWithHarry**: "Python Complete Course in Hindi" (12 hours)
- **Corey Schafer**: "Python Tutorial for Beginners" (4 hours)

Skip: Decorators, generators, metaclasses (not needed for AI)

Week 10: LLM Fundamentals + Prompt Engineering

Topic	What to Learn	Practice
LLM Basics	Transformers, attention, tokenization	Read 5 articles

Prompt Engineering	Zero-shot, few-shot, chain-of-thought	Write 20 prompts
Parameters	Temperature, max_tokens, top_p	Experiment with 5 prompts
OpenAI API	chat completions, streaming	Build simple chatbot
Gemini API	Free alternative, same concepts	Replace OpenAI with Gemini

YouTube Resources:

- **FreeCodeCamp:** "Generative AI Full Course" (8 hours)
- **CodeWithHarry:** "GenAI Complete Course in Hindi" (10 hours)
- **Nicholas Renotte:** "Prompt Engineering Tutorial" (3 hours)

Project 3: Chat Assistant CLI (Python)

- Use OpenAI/Gemini API
 - Conversation history
 - Temperature control
 - Stream responses
 - Save chat history to file
-

Week 11: AI Integration

Topic	What to Learn	Practice
API Integration	REST API, async, error handling	Connect 3 AI APIs
Image Generation	DALL-E, Stable Diffusion	Generate 10 images
Text-to-Speech	Whisper, Audio APIs	Convert 5 texts to audio
Speech-to-Text	Whisper API, transcription	Transcribe 5 audios

YouTube Resources:

- **Nicholas Renotte:** "AI Integration Tutorial" (4 hours)
- **FreeCodeCamp:** "OpenAI API Full Course" (3 hours)
- **CodeWithHarry:** "AI APIs Complete Course" (5 hours)

Project 4: AI Image Generator (Full Stack)

- React frontend

- Node.js backend
 - DALL-E API integration
 - Save images to local folder
 - Display in browser
-

Week 12: GenAI + Full Stack Integration

Topic	What to Learn	Practice
FastAPI	Python backend, routes, async	Build 2 AI APIs
API Gateway	Node.js routes to FastAPI	Connect 2 services
Error Handling	Try-except, status codes	Handle errors in 2 APIs
Logging	print, logging module	Add logging to 2 apps

YouTube Resources:

- **FreeCodeCamp**: "FastAPI Full Course" (4 hours)
- **CodeWithHarry**: "FastAPI Complete Course in Hindi" (3 hours)
- **FastAPI**: "Official FastAPI Tutorial" (2 hours)

Project 3 Update: Add Web Frontend

- Convert CLI to React frontend
 - Node.js backend talks to FastAPI
 - FastAPI calls OpenAI/Gemini
 - Full-stack AI chat app
-

MONTH 4: RAG + LangChain + Vector Databases (Weeks 13–16)

Week 13: LangChain Basics

Topic	What to Learn	Practice
Chains	LLMChain, SequentialChain	Build 3 chains

Models	Chat models, text models	Use 2 models
Prompts	Prompt templates, formatting	Create 5 prompts
Memory	ConversationBufferMemory	Add memory to 2 chatbots
Agents	AgentExecutor, tools	Build 1 AI agent

YouTube Resources:

- **FreeCodeCamp:** "LangChain Full Course" (6 hours)
 - **Nicholas Renotte:** "LangChain Tutorial" (4 hours)
 - **CodeWithHarry:** "LangChain Complete Course in Hindi" (5 hours)
 - **LangChain:** "Official LangChain Tutorial" (2 hours)
-

Week 14: Document Loaders + Text Splitters

Topic	What to Learn	Practice
Loaders	PDF, TXT, CSV, web	Load 5 documents
Splitters	Recursive, fixed-size, semantic	Split 3 large documents
Embeddings	OpenAI embeddings, HuggingFace	Create embeddings for 50 texts

YouTube Resources:

- **Nicholas Renotte:** "LangChain Document Loaders" (2 hours)
 - **LangChain:** "Document Loaders Tutorial" (1 hour)
-

Week 15: Vector Databases

Topic	What to Learn	Practice
Pinecone	Setup, index, upsert, query	Setup free tier, store 100 docs
MongoDB Atlas	Vector search, indexes	Setup vector search
Chroma	Local vector DB, persistence	Use for 1 project

Similarity Search	KNN, cosine similarity, top-k	Search 100 documents
--------------------------	-------------------------------	----------------------

YouTube Resources:

- **FreeCodeCamp:** "Vector Databases Full Course" (3 hours)
- **Pinecone:** "Pinecone Tutorial" (2 hours)
- **Nicholas Renotte:** "Pinecone Tutorial" (2 hours)
- **MongoDB:** "MongoDB Vector Search" (2 hours)

Week 16: RAG Pipeline

Topic	What to Learn	Practice
RAG Basics	Chunking → Embedding → Retrieval → Generation	Build full RAG system
Retrievers	Vector retriever, hybrid search	Use 2 retrievers
Optimization	Re-ranking, filtering, hybrid	Optimize 2 RAG systems
Evaluation	Accuracy, relevance, faithfulness	Evaluate 1 RAG system

YouTube Resources:

- **FreeCodeCamp:** "RAG Full Course" (5 hours)
- **Nicholas Renotte:** "RAG Tutorial" (4 hours)
- **CodeWithHarry:** "RAG Complete Course in Hindi" (4 hours)

Project 5: AI Resume Analyzer (RAG)

- React frontend (upload PDF)
- Node.js backend (Express)
- Python FastAPI (LangChain + RAG)
- PostgreSQL (user data)
- Pinecone (embeddings for skills)
- OpenAI API (generation)
- Features:
 - Upload PDF resume
 - Extract text with PyPDF2
 - Create embeddings for skills
 - Store in Pinecone
 - Query: "Does this resume have Python?"
 - Generate analysis with OpenAI

- Score: 0–100 based on job match
 - **1000+ lines of code**
 - **Deploy on AWS** (Week 19–20)
-

MONTH 5: Cloud + Docker + CI/CD + Capstone Projects (Weeks 17–20)

Week 17: Docker + Containerization

Topic	What to Learn	Practice
Basics	docker run, ps, images, rm	Run 5 containers
Dockerfile	FROM, RUN, CMD, EXPOSE, COPY	Create Dockerfile for 3 apps
Docker Compose	Services, networks, volumes	Build docker-compose for 2 apps
Docker Hub	Push, pull, login, tags	Push 2 images to Docker Hub

YouTube Resources:

- **FreeCodeCamp:** "Docker Full Course" (5 hours)
- **CodeWithHarry:** "Docker Complete Course in Hindi" (4 hours)
- **TechWorld with Nana:** "Docker Tutorial for Beginners" (6 hours)
- **Docker:** "Official Docker Tutorial" (3 hours)

Project 5 Update: Dockerize AI Resume Analyzer

- Create a Dockerfile for the React frontend
 - Create a Dockerfile for Node.js backend
 - Create Dockerfile for FastAPI (Python)
 - docker-compose for React + Node + FastAPI + PostgreSQL + Pinecone
 - Push all images to Docker Hub
-

Week 18: AWS + Cloud Deployment

Topic	What to Learn	Practice
EC2	Create instance, SSH, security groups	Deploy 2 apps on EC2
S3	Buckets, upload, policies, CORS	Upload files to S3
Lambda	Functions, triggers, events	Create 3 Lambda functions
RDS	PostgreSQL on AWS	Setup production database
IAM	Users, roles, policies	Secure 2 apps with IAM

YouTube Resources:

- **FreeCodeCamp**: "AWS Full Course for Beginners" (12 hours)
- **CodeWithHarry**: "AWS Complete Course in Hindi" (10 hours)
- **AWS**: "AWS Official Tutorial" (5 hours)
- **TechWithTim**: "AWS for Beginners" (4 hours)

Project 5 Deploy: AI Resume Analyzer on AWS

- Deploy backend on EC2
 - Store PDF files on S3
 - Use RDS for production PostgreSQL
 - Setup Lambda for image processing (optional)
 - Configure IAM for security
-

Week 19: GitHub Actions + CI/CD

Topic	What to Learn	Practice
Basics	Triggers, workflows, jobs, steps	Create 3 workflows
CI	Build, test, lint	Setup CI for 2 apps
CD	Deploy to EC2, Docker Hub, Netlify	Setup CD for 2 apps

Secrets	GitHub Secrets, environment vars	Secure 3 API keys
----------------	----------------------------------	-------------------

YouTube Resources:

- **FreeCodeCamp:** "GitHub Actions Full Course" (3 hours)
- **CodeWithHarry:** "GitHub Actions Complete Course" (2 hours)
- **GitHub:** "Official GitHub Actions Tutorial" (2 hours)
- **Jim Petterson:** "CI/CD with GitHub Actions" (3 hours)

Project 5 Final: Auto-Deploy AI Resume Analyser

- Create .github/workflows/deploy.yml
- Auto-run tests on PR
- Build Docker images
- Push to Docker Hub
- Deploy to EC2 automatically

Week 20: Capstone Project 2 + Portfolio

Project 6: AI Customer Support Agent

Features	Tech Stack
Multi-turn conversation (memory)	LangChain Memory
RAG: Company knowledge base (100+ docs)	Pinecone + LangChain
AI Agents: Auto-route to human	LangGraph
Dashboard: Track conversations	React
Analytics: Response time, satisfaction	PostgreSQL
Production-ready: Logging, error handling	Sentry

Tech Stack:

- Next.js (frontend + API routes)
- Node.js + Express (web backend)
- FastAPI (Python backend for AI)
- PostgreSQL (users + conversations)
- Pinecone (knowledge base)
- LangGraph (AI agents)
- Docker + AWS ECS

- GitHub Actions (auto-deploy)
- Sentry (error tracking)

Code: 1200+ lines

Deploy: Production with monitoring

FINAL PROJECT LIST

#	Project	Tech Stack	Lines of Code
1	E-commerce Backend (PostgreSQL + Redis)	Node.js, Express, PostgreSQL, Prisma, Redis	800+
2	Social Media App (Full Stack)	React, Node.js, Express, PostgreSQL, JWT	1000+
3	Chat Assistant CLI + Web	Python, FastAPI, OpenAI/Gemini, React	400+
4	AI Image Generator	React, Node.js, DALL-E API	300+
5	AI Resume Analyzer (RAG)	React, Node.js, FastAPI, PostgreSQL, Pinecone, LangChain, OpenAI	1000+
6	AI Customer Support Agent	Next.js, Node.js, FastAPI, PostgreSQL, Pinecone, LangGraph	1200+

Total: 6 Projects | **4700+ Lines of Code**

COMPLETE TECH STACK SUMMARY

Category	Technologies
Frontend	React, Next.js
Backend (Web)	Node.js, Express.js
Backend (AI)	FastAPI (Python)
Database	PostgreSQL, Prisma ORM, Redis
Vector DB	Pinecone

GenAI	OpenAI API, Gemini API
RAG	LangChain, LangGraph, Embeddings, Text Splitters
DevOps	Docker, Docker Compose, GitHub Actions
Cloud	AWS (EC2, S3, Lambda, RDS, IAM)
Tools	Git, GitHub, VS Code, Postman, Sentry
DSA	C++ (LeetCode)



BEST FREE YouTube RESOURCES (All-in-One)

Topic	Best Channel	Course Name	Duration
PostgreSQL	FreeCodeCamp	PostgreSQL Full Course	5 hrs
Prisma ORM	Prisma	Prisma ORM Tutorial	3 hrs
Redis	FreeCodeCamp	Redis Full Course	2 hrs
JWT Auth	FreeCodeCamp	JWT Authentication Full Course	2 hrs
DSA	CodeWithHarry	DSA Complete Course in Hindi	15 hrs
DSA Problems	NeetCode	DSA Roadmap + Problems	10 hrs
Python	FreeCodeCamp	Python for Beginners	4 hrs
GenAI	FreeCodeCamp	Generative AI Full Course	8 hrs
Prompt Engineering	Nicholas Renotte	Prompt Engineering Tutorial	3 hrs
OpenAI API	FreeCodeCamp	OpenAI API Full Course	3 hours
FastAPI	FreeCodeCamp	FastAPI Full Course	4 hrs
LangChain	FreeCodeCamp	LangChain Full Course	6 hrs
RAG	FreeCodeCamp	RAG Full Course	5 hrs
Pinecone	Pinecone	Pinecone Tutorial	2 hrs

Docker	TechWorld with Nana	Docker Tutorial for Beginners	6 hrs
AWS	FreeCodeCamp	AWS Full Course for Beginners	12 hrs
GitHub Actions	FreeCodeCamp	GitHub Actions Full Course	3 hrs
System Design	Gaurav Sen	System Design Basics	3 hrs

✓ FINAL CHECKLIST FOR INDUSTRY-READY

- ✓ **200+ LeetCode** problems solved (C++)
- ✓ **6 Production Projects** (4700+ lines of code)
- ✓ **Full Stack**: React + Node.js + PostgreSQL + Redis
- ✓ **GenAI**: OpenAI + Gemini + FastAPI
- ✓ **RAG**: LangChain + LangGraph + Embeddings + Pinecone
- ✓ **DevOps**: Docker + Docker Compose + GitHub Actions
- ✓ **Cloud**: AWS (EC2 + S3 + Lambda + RDS + IAM)
- ✓ **Live Demos**: All 6 projects deployed publicly
- ✓ **GitHub**: 6 public repos with README + screenshots
- ✓ **Portfolio**: Professional website showcasing projects
- ✓ **Resume**: 1-page resume with skills + projects

🎯 EXPECTED PACKAGE AFTER THIS ROADMAP

Profile	Fresher Package	3-Year Package	5-Year Package
Following This Roadmap	₹10–16 LPA	₹20–30 LPA	₹30–45+ LPA
With 250+ LC + 6 Projects + Deployed	₹12–18 LPA	₹25–35 LPA	₹35–50+ LPA

UPDATED ROADMAP FOR YOUR CASE (Know MERN + DSA in C++)

What You Already Have:

-  **MERN** (React + Node + MongoDB)
-  **DSA in C++** (keep it!)

What You Need to Add:

1. **PostgreSQL** (replace MongoDB for production)
2. **Redis** (caching + Pub/Sub)
3. **Python** (2 weeks - only for AI)
4. **FastAPI** (Python backend for AI)
5. **LangChain + LangGraph** (RAG + AI Agents)
6. **Pinecone** (Vector DB)
7. **OpenAI/Gemini APIs** (LLMs)
8. **Docker** (containerization)
9. **AWS** (cloud deployment)
10. **GitHub Actions** (CI/CD)

Time Allocation:

- **Month 1:** PostgreSQL + Redis + Security (keep React/Node)
- **Month 2:** DSA (C++) + System Design (keep React/Node)
- **Month 3:** **Python (2 weeks)** + GenAI APIs
- **Month 4:** **LangChain + RAG + Pinecone** (Python)
- **Month 5:** Docker + AWS + CI/CD + Projects (React + Node + Python)

Python Learning Time: Only **2 weeks** (Month 3, Week 1) for basics

Start today. Follow week-by-week. Build all 6 projects. Deploy everything. You'll be Full Stack + Cloud + GenAI + RAG Engineer in 5 months. 

IN-DEPTH

Full Stack AI Engineer Roadmap

Beginner from Scratch → Industry-Ready in 9 Months

 OVERALL TIMELINE

Phase	Duration	Focus
Phase 1	Month 1–2	Programming Fundamentals + Web Basics
Phase 2	Month 3–4	Full Stack Development (React + Node)
Phase 3	Month 5	DSA + Logic Building
Phase 4	Month 6	GenAI Basics (LLMs, APIs, Prompt Engineering)
Phase 5	Month 7	Advanced GenAI (LangChain, RAG, Vector DBs)
Phase 6	Month 8	System Design + DevOps (Docker, AWS, CI/CD)
Phase 7	Month 9	Capstone Projects + Placement Prep

Total: 9 Months | Daily Hours: 4–6 hours

 PHASE 1: Programming Fundamentals + Web Basics (Month 1–2)

Week 1–2: Python Fundamentals (*Essential for AI*)

Topic	What to Learn	Practice
Variables & Data Types	int, float, string, bool, list, dict, tuple, set	Write 10 programs using each type
Operators	Arithmetic, comparison, logical, assignment	Solve 15 math problems
Conditionals	if, elif, else, nested if	Write 10 decision-making programs
Loops	for, while, break, continue, nested loops	Write 15 iteration programs
Functions	def, parameters, return, scope, lambda	Create 10 reusable functions
Error Handling	try, except, finally, raise	Handle errors in 5 programs

YouTube Resources:

- **Corey Schafer:** "Python Tutorial for Beginners" (4 hours)
- **CodeWithHarry:** "Python Complete Course in Hindi" (12 hours)
- **FreeCodeCamp:** "Python for Beginners" (4 hours)

Mini-Project 1: Calculator App (CLI)

- Add, subtract, multiply, divide
- Handle division by zero
- Save history to file

Week 3–4: JavaScript Fundamentals

Topic	What to Learn	Practice
Variables & Types	let, const, var, types, typeof	Write 10 programs
Operators	All JS operators	Solve 15 problems
Conditionals	if, else, switch	Write 10 logic programs

Loops	for, while, forEach, map, filter	Write 15 array programs
Functions	function, arrow, callback, closure	Create 10 functions
ES6+	Template literals, destructuring, and spread	Modern JS syntax

YouTube Resources:

- **FreeCodeCamp:** "JavaScript Tutorial for Beginners" (3 hours)
- **CodeWithHarry:** "JavaScript Complete Course" (10 hours)
- **The Net Ninja:** "JavaScript Tutorial" (5 hours)

Mini-Project 2: Todo List (CLI)

- Add, delete, mark complete
- Save to JSON file

Week 5–6: HTML & CSS

Topic	What to Learn	Practice
HTML Basics	Tags, attributes, elements, forms	Build 5 simple pages
HTML Advanced	Div, section, header, footer, navigation	Create 3 webpage layouts
CSS Basics	Colors, fonts, padding, margin, border	Style 5 pages
CSS Layouts	Flexbox, Grid, position	Create 3 responsive layouts
CSS Advanced	Animations, transitions, media queries	Make 2 animated pages

YouTube Resources:

- **FreeCodeCamp:** "HTML & CSS Full Course" (5 hours)
- **CodeWithHarry:** "HTML CSS Complete Course" (6 hours)
- **Kevin Powell:** "CSS for Beginners" (4 hours)

Mini-Project 3: Personal Portfolio Website

- Home, About, Projects, Contact sections
 - Responsive design
 - Simple animations
-

Week 7–8: Git & GitHub + Basic Logic

Topic	What to Learn	Practice
Git Basics	init, add, commit, push, pull, clone	Create 5 repositories
Git Advanced	Branch, merge, rebase, conflict resolution	Merge 3 branches
GitHub	Repo, README, contributions, projects	Create 3 public repos
Logic Building	Patterns, arrays, strings, basic DP	Solve 50 HackerRank problems

YouTube Resources:

- **FreeCodeCamp**: "Git & GitHub Full Course" (2 hours)
- **CodeWithHarry**: "Git GitHub Complete Course" (3 hours)
- **Hitesh Choudhary**: "Git & GitHub in Hindi" (2 hours)

Mini-Project 4: Upload Portfolio to GitHub

- Push all previous projects
 - Write detailed README
 - Get first GitHub contribution
-

PHASE 2: Full Stack Development (Month 3–4)

Week 9–12: React.js (Advanced Frontend)

Topic	What to Learn	Practice
React Basics	Components, props, state, render	Build 3 component-based apps

Hooks	useState, useEffect, useContext, useRef	Create 5 hook-based apps
Advanced Hooks	useReducer, useMemo, useCallback, custom hooks	Optimize 3 apps
React Router	Routes, parameters, navigation	Build 3 multi-page apps
API Calls	fetch, axios, async, error handling	Connect 3 APIs
State Management	Context API, Redux, Zustand	Manage state in 2 apps
Forms	Formik, React Hook Form, validation	Build 3 form apps
Performance	Lazy loading, memoisation, code splitting	Optimize 2 apps

YouTube Resources:

- **FreeCodeCamp:** "React Tutorial for Beginners" (12 hours)
- **CodeWithHarry:** "React Complete Course" (15 hours)
- **The Net Ninja:** "React Tutorial" (10 hours)
- **PedroTech:** "React Hooks Explained" (3 hours)

Project 1: E-commerce Frontend

- Product listing with search/filter
- Product details page
- Shopping cart with localStorage
- User authentication UI
- Responsive design

Week 13–16: Node.js + Express (Backend)

Topic	What to Learn	Practice
Node Basics	Runtime, modules, fs, path, events	Build 3 CLI tools
Express Basics	Server, routes, middleware, request/response	Create 3 REST APIs
Express Advanced	Router, error handling, validation	Build 5 APIs

Authentication	JWT, bcrypt, sessions, OAuth	Implement auth in 2 apps
Databases	MongoDB (Mongoose), PostgreSQL (Prisma)	Connect 3 databases
API Design	REST, status codes, versioning, documentation	Design 3 APIs
Security	CORS, Helmet, rate limiting, sanitisation	Secure 2 APIs
File Upload	Multer, S3, Cloudinary	Upload files in 2 apps

YouTube Resources:

- **FreeCodeCamp:** "Node.js & Express Full Course" (10 hours)
- **CodeWithHarry:** "Node.js Complete Course" (12 hours)
- **The Net Ninja:** "Node.js Tutorial" (8 hours)
- **Web Dev Simplified:** "Express.js Tutorial" (4 hours)

Project 2: E-commerce Backend API

- User registration/login (JWT + bcrypt)
- Product CRUD operations
- Shopping cart management
- Order placement
- Payment integration (Stripe/Razorpay sandbox)
- Email notifications (Nodemailer)
- RESTful API with 15+ endpoints

Week 17–18: Database Mastery

Topic	What to Learn	Practice
MongoDB	Schema, models, relationships, indexing	Design 3 databases
PostgreSQL	Tables, JOINS, constraints, transactions	Build 3 relational DBs
Prisma ORM	Schema, queries, migrations, types	Connect Prisma to 3 apps
Advanced Queries	Aggregation, filtering, pagination	Optimize 5 queries

Caching	Redis, cache strategies, TTL	Cache 2 APIs
---------	------------------------------	--------------

YouTube Resources:

- **FreeCodeCamp:** "MongoDB Full Course" (6 hours)
- **FreeCodeCamp:** "PostgreSQL Full Course" (5 hours)
- **CodeWithHarry:** "MongoDB Complete Course" (5 hours)
- **Prisma:** "Prisma ORM Tutorial" (3 hours)

Mini-Project 5: Blog Platform with Database

- Users, posts, comments (PostgreSQL + Prisma)
- Search and filter posts
- Pagination
- Redis caching for popular posts

PHASE 3: DSA + Logic Building (Month 5)

Week 19–20: DSA Basics

Topic	What to Learn	LeetCode Problems
Arrays	Traversal, searching, sorting, 2-pointer	Two Sum, Best Time to Buy Stock, Contains Duplicate
Strings	Palindrome, reversal, frequency	Valid Palindrome, Reverse Words, Valid Anagram
Hashing	HashMap, HashSet, frequency counting	Group Anagrams, Top K Frequent Elements
Linked List	Reverse, merge, cycle detection	Reverse LL, Merge Two Sorted Lists, Linked List Cycle

YouTube Resources:

- **FreeCodeCamp:** "DSA Full Course" (12 hours)
- **CodeWithHarry:** "DSA Complete Course in Hindi" (15 hours)
- **NeetCode:** "DSA Roadmap" (2 hours)
- **TakeUForward:** "DSA in 30 Days" (10 hours)

Target: 100 LeetCode (Easy: 60, Medium: 40)

Week 21–22: DSA Intermediate

Topic	What to Learn	LeetCode Problems
Stacks	Push, pop, parentheses, monotonic stack	Valid Parentheses, Min Stack, Daily Temperatures
Trees	DFS, BFS, traversal, BST	Maximum Depth, Invert Tree, Level Order, Validate BST
Recursion	Base case, recursive tree, backtracking	Factorial, Fibonacci, Subsets, Permutations

YouTube Resources:

- **NeetCode:** "Tree Problems Explained" (5 hours)
- **TakeUForward:** "Recursion Complete Course" (6 hours)
- **CodeWithHarry:** "Trees Complete Course" (5 hours)

Target: 200 LeetCode (Easy: 80, Medium: 100, Hard: 20)

Week 23–24: DSA Advanced

Topic	What to Learn	LeetCode Problems
Graphs	BFS, DFS, adjacency, shortest path	Number of Islands, Clone Graph, Course Schedule
Dynamic Programming	1D, 2D, memoization, tabulation	Climbing Stairs, House Robber, Coin Change, LCS
Heaps	Min heap, max heap, priority queue	Kth Largest Element, Merge K Sorted Lists

YouTube Resources:

- **NeetCode:** "Graph Problems Explained" (6 hours)
- **TakeUForward:** "DP Complete Course" (8 hours)
- **CodeWithHarry:** "Graph Complete Course" (5 hours)

Target: 300 LeetCode (Easy: 80, Medium: 180, Hard: 40)

PHASE 4: GenAI Basics (Month 6)

Week 25–26: LLM Fundamentals + Prompt Engineering

Topic	What to Learn	Practice
LLM Basics	Transformers, attention, tokenisation, embeddings	Read 5 articles on transformers
Prompt Engineering	Zero-shot, few-shot, chain-of-thought, temperature	Write 20 prompts for different tasks
OpenAI API	Chat completions, parameters, streaming	Build a simple chatbot
Gemini API	Free alternative, same concepts	Replace OpenAI with Gemini
HuggingFace	Models, pipelines, tokenisers	Use 5 pre-trained models

YouTube Resources:

- **FreeCodeCamp**: "Generative AI Full Course" (8 hours)
- **CodeWithHarry**: "GenAI Complete Course in Hindi" (10 hours)
- **Nicholas Renotte**: "Prompt Engineering Tutorial" (3 hours)
- **HuggingFace**: "HuggingFace Tutorial" (2 hours)

Project 3: Chat Assistant CLI

- Use OpenAI/Gemini API
- Conversation history
- Temperature control
- Stream responses
- Save chat history to file

Week 27–28: AI Integration

Topic	What to Learn	Practice
API Integration	REST API, async, error handling	Connect 3 AI APIs
Image Generation	DALL-E, Stable Diffusion	Generate 10 images

Text-to-Speech	Whisper, Audio APIs	Convert 5 texts to audio
Speech-to-Text	Whisper API, transcription	Transcribe 5 audios
AI Models	Fine-tuning basics, LoRA, QLoRA	Try 1 pre-trained model

YouTube Resources:

- **Nicholas Renotte:** "AI Integration Tutorial" (4 hours)
- **FreeCodeCamp:** "OpenAI API Full Course" (3 hours)
- **CodeWithHarry:** "AI APIs Complete Course" (5 hours)

Mini-Project 6: AI Image Generator

- Input text prompt
- Generate image with DALL-E
- Save to local folder
- Display in browser

PHASE 5: Advanced GenAI (Month 7)

Week 29–30: LangChain

Topic	What to Learn	Practice
LangChain Basics	Chains, models, prompts, output parsers	Build 3 simple chains
Memory	ConversationBufferMemory, RedisMemory	Add memory to 2 chatbots
Agents	AgentExecutor, tools, ReAct	Build 2 AI agents
Document Loaders	PDF, TXT, CSV, web	Load 5 documents
Text Splitters	Recursive, fixed-size, semantic	Split 3 large documents

YouTube Resources:

- **FreeCodeCamp:** "LangChain Full Course" (6 hours)
- **Nicholas Renotte:** "LangChain Tutorial" (4 hours)
- **CodeWithHarry:** "LangChain Complete Course in Hindi" (5 hours)
- **LangChain:** "LangChain Official Tutorial" (2 hours)

Mini-Project 7: AI Document Q&A

- Load PDF document
 - Create LangChain chain
 - Ask questions about document
 - Get answers with context
-

Week 31–32: RAG + Vector Databases

Topic	What to Learn	Practice
Embeddings	OpenAI embeddings, HuggingFace	Create embeddings for 50 texts
Vector DBs	Pinecone, MongoDB Atlas, Weaviate, Chroma	Set up Pinecone (free tier)
Similarity Search	KNN, cosine similarity, top-k	Search 100 documents
RAG Pipeline	Chunking → Embedding → Retrieval → Generation	Build a full RAG system
RAG Optimization	Hybrid search, re-ranking, filtering	Optimise 2 RAG systems

YouTube Resources:

- **FreeCodeCamp**: "RAG Full Course" (5 hours)
- **Nicholas Renotte**: "RAG Tutorial" (4 hours)
- **CodeWithHarry**: "RAG Complete Course in Hindi" (4 hours)
- **Pinecone**: "Pinecone Tutorial" (2 hours)

Project 4: AI Resume Analyzer (RAG)

- Upload PDF resume
- Extract text with PyPDF2
- Create embeddings for skills
- Store in Pinecone
- Query: "Does this resume have Python?"
- Generate analysis with OpenAI
- Score: 0–100 based on job match
- Frontend: React
- Backend: Node.js + Express
- Database: PostgreSQL + Pinecone

PHASE 6: System Design + DevOps (Month 8)

Week 33–34: Docker + Containerization

Topic	What to Learn	Practice
Docker Basics	Images, containers, run, ps, rm	Run 5 containers
Dockerfile	FROM, RUN, CMD, EXPOSE, COPY	Create Dockerfile for 3 apps
Docker Compose	Services, networks, volumes	Build docker-compose for 2 apps
Docker Hub	Push, pull, login, tags	Push 2 images to Docker Hub
Best Practices	Multi-stage builds, optimisation	Optimize 2 Dockerfiles

YouTube Resources:

- **FreeCodeCamp:** "Docker Full Course" (5 hours)
- **CodeWithHarry:** "Docker Complete Course in Hindi" (4 hours)
- **TechWorld with Nana:** "Docker Tutorial" (6 hours)
- **Docker:** "Official Docker Tutorial" (3 hours)

Project 5: Dockerize E-commerce App

- Create Dockerfile for React frontend
- Create Dockerfile for Node backend
- docker-compose for React + Node + PostgreSQL
- Push all images to Docker Hub

Week 35–36: AWS + Cloud Deployment

Topic	What to Learn	Practice
EC2	Create instance, SSH, security groups	Deploy 2 apps on EC2
S3	Buckets, upload, policies, CORS	Upload files to S3

Lambda	Functions, triggers, events	Create 3 Lambda functions
RDS	PostgreSQL on AWS	Setup production database
IAM	Users, roles, policies	Secure 2 apps with IAM
CloudFront	CDN, distribution, caching	Setup CDN for 1 app

YouTube Resources:

- **FreeCodeCamp:** "AWS Full Course" (12 hours)
- **CodeWithHarry:** "AWS Complete Course in Hindi" (10 hours)
- **AWS:** "AWS Official Tutorial" (5 hours)
- **TechWithTim:** "AWS for Beginners" (4 hours)

Project 6: Deploy AI Resume Analyzer on AWS

- Deploy backend on EC2
- Store files on S3
- Use RDS for production DB
- Setup Lambda for image processing
- Configure IAM for security

Week 37–38: GitHub Actions + CI/CD

Topic	What to Learn	Practice
GitHub Actions Basics	Triggers, workflows, jobs, steps	Create 3 workflows
CI (Continuous Integration)	Build, test, lint	Setup CI for 2 apps
CD (Continuous Deployment)	Deploy to EC2, Docker Hub, Netlify	Setup CD for 2 apps
Secrets	GitHub Secrets, environment vars	Secure 3 API keys
Best Practices	Optimization, caching, parallel jobs	Optimize 2 workflows

YouTube Resources:

- **FreeCodeCamp:** "GitHub Actions Full Course" (3 hours)
- **CodeWithHarry:** "GitHub Actions Complete Course" (2 hours)

- **GitHub:** "Official GitHub Actions Tutorial" (2 hours)
- **Jim Petterson:** "CI/CD with GitHub Actions" (3 hours)

Project 7: Auto-Deploy E-commerce App

- Create .github/workflows/deploy.yml
 - Auto-run tests on PR
 - Build Docker image
 - Push to Docker Hub
 - Deploy to EC2 automatically
-

PHASE 7: Capstone Projects + Placement Prep (Month 9)

Week 39–40: Capstone Project 1 – AI Interview Assistant

Features:

- Real-time chat (WebSocket + Redis Pub/Sub)
- Interview questions from LangChain
- RAG: Knowledge base of tech topics
- Voice input/output (optional)
- Session history (PostgreSQL)
- Rate limiting + auth
- Dashboard for analytics

Tech Stack:

- React + Socket.io (frontend)
- Node.js + WebSocket (backend)
- Redis (caching + Pub/Sub)
- PostgreSQL (sessions)
- LangChain (AI)
- Gemini API (free)
- Docker + AWS
- GitHub Actions (CI/CD)

Code: 1000+ lines

Deploy: Live with WebSocket support

Week 41–42: Capstone Project 2 – AI Customer Support Agent

Features:

- Multi-turn conversation (memory)
- RAG: Company knowledge base (100+ docs)
- AI Agents: Auto-route to human if needed
- Dashboard: Track conversations
- Analytics: Response time, satisfaction
- Production-ready: Logging, error handling
- Monitoring: Sentry integration

Tech Stack:

- Next.js (frontend + API routes)
- FastAPI (Python backend for AI)
- PostgreSQL (users + conversations)
- Pinecone (knowledge base)
- LangGraph (AI agents)
- Docker + AWS ECS
- GitHub Actions (auto-deploy)
- Sentry (error tracking)

Code: 1200+ lines

Deploy: Production with monitoring

Week 43–44: Placement Prep + Portfolio

Activity	What to Do	Time
DSA Review	Review 300 LC problems, solve 20 new	2 weeks
CS Fundamentals	OOP, DBMS, OS, CN, SQL (100 questions)	1 week
System Design	Low-level design for 5 apps	1 week
Portfolio Website	Showcase all 7 projects with screenshots	1 week
LinkedIn Optimization	Update profile, add projects, connect	3 days

Resume Writing	1-page resume with projects + skills	3 days
Mock Interviews	10 mock interviews (Pramp, interviewing.io)	1 week

YouTube Resources:

- **NeetCode:** "Interview Prep Roadmap" (2 hours)
- **CodeWithHarry:** "Placement Prep Complete Course" (8 hours)
- **TakeUForward:** "System Design for Interviews" (5 hours)
- **Pramp:** "Mock Interview Tips" (1 hour)

 FINAL PROJECT LIST

#	Project	Tech Stack	Lines of Code
1	E-commerce Frontend	React, Redux, REST API	600+
2	E-commerce Backend	Node.js, Express, MongoDB/PostgreSQL	800+
3	Chat Assistant CLI	Python, OpenAI/Gemini API	200+
4	AI Resume Analyzer	React, Node, PostgreSQL, Pinecone, OpenAI	1000+
5	Dockerized E-commerce	React, Node, PostgreSQL, Docker	1400+ (combined)
6	AI Interview Assistant	React, Socket.io, Redis, LangChain, Gemini	1000+
7	AI Customer Support Agent	Next.js, FastAPI, PostgreSQL, Pinecone, LangGraph	1200+

Total: 7 Projects | 5000+ Lines of Code



COMPLETE TECH STACK SUMMARY

Category	Technologies
Programming	Python, JavaScript
Frontend	React, Redux, Next.js, Socket.io
Backend	Node.js, Express, FastAPI
Database	MongoDB, PostgreSQL, Prisma ORM
Caching	Redis
AI/GenAI	OpenAI API, Gemini API, LangChain, LangGraph
RAG	Embeddings, Vector DBs (Pinecone, MongoDB Atlas), Text Splitters
DevOps	Docker, Docker Compose, GitHub Actions
Cloud	AWS (EC2, S3, Lambda, RDS, IAM, CloudFront)
Tools	Git, GitHub, VS Code, Postman, Sentry



WEEKLY CHECKLIST

Week	DSA Target	Dev Target	GenAI Target	DevOps Target
1–2	20 LC	Python Complete	—	—
3–4	40 LC	JavaScript Complete	—	—
5–6	60 LC	HTML/CSS + Portfolio	—	—
7–8	80 LC	Git + GitHub + Logic	—	—
9–12	120 LC	React Complete	—	—
13–16	160 LC	Node + Express + Auth	—	—
17–18	200 LC	Database Mastery	—	—
19–20	220 LC	—	DSA Basics	—
21–22	260 LC	—	DSA Intermediate	—

23–24	300 LC	—	DSA Advanced	—
25–26	Review	—	LLM + Prompt + APIs	—
27–28	Review	—	AI Integration	—
29–30	Review	—	LangChain	—
31–32	Review	—	RAG + Vector DBs	—
33–34	Review	Dockerize Apps	—	Docker Complete
35–36	Review	Deploy on AWS	—	AWS Complete
37–38	Review	CI/CD Setup	—	GitHub Actions
39–40	Review	Project 6	Project 6 AI	Project 6 Deploy
41–42	Review	Project 7	Project 7 AI	Project 7 Deploy
43–44	320 LC	Portfolio	—	—

BEST FREE Youtube RESOURCES (All-in-One)

Topic	Best Channel	Course Name	Duration
Python	CodeWithHarry	Python Complete Course in Hindi	12 hrs
JavaScript	CodeWithHarry	JavaScript Complete Course in Hindi	10 hrs
HTML/CSS	FreeCodeCamp	HTML & CSS Full Course	5 hrs
React	FreeCodeCamp	React Tutorial for Beginners	12 hrs
Node.js	CodeWithHarry	Node.js Complete Course in Hindi	12 hrs
MongoDB	FreeCodeCamp	MongoDB Full Course	6 hrs
PostgreSQL	FreeCodeCamp	PostgreSQL Full Course	5 hrs
DSA	CodeWithHarry	DSA Complete Course in Hindi	15 hrs
DSA Problems	NeetCode	DSA Roadmap + Problems	10 hrs
GenAI Basics	FreeCodeCamp	Generative AI Full Course	8 hrs

Prompt Engineering	Nicholas Renotte	Prompt Engineering Tutorial	3 hrs
OpenAI API	FreeCodeCamp	OpenAI API Full Course	3 hrs
LangChain	FreeCodeCamp	LangChain Full Course	6 hrs
RAG	FreeCodeCamp	RAG Full Course	5 hrs
Pinecone	Pinecone	Pinecone Tutorial	2 hrs
Docker	TechWorld with Nana	Docker Tutorial for Beginners	6 hrs
AWS	FreeCodeCamp	AWS Full Course for Beginners	12 hrs
GitHub Actions	FreeCodeCamp	GitHub Actions Full Course	3 hrs
System Design	TakeUForward	System Design for Interviews	5 hrs

Bonus: All channels have playlists – watch in order!

FINAL CHECKLIST FOR INDUSTRY-READY

-  **300+ LeetCode** problems solved
 -  **7 Production Projects** (5000+ lines of code)
 -  **Full Stack:** React + Node + PostgreSQL + Redis
 -  **GenAI:** OpenAI + Gemini + LangChain + LangGraph
 -  **RAG:** Embeddings + Vector DBs + Retrieval + Generation
 -  **DevOps:** Docker + Docker Compose + GitHub Actions
 -  **Cloud:** AWS (EC2 + S3 + Lambda + RDS + IAM)
 -  **Live Demos:** All 7 projects deployed publicly
 -  **GitHub:** 7 public repos with README + screenshots
 -  **Portfolio:** Professional website showcasing projects
 -  **Resume:** 1-page resume with skills + projects
 -  **LinkedIn:** Optimized profile with 100+ connections
-



EXPECTED PACKAGE AFTER THIS ROADMAP

Profile	Fresher Package	3-Year Package	5-Year Package
Following This Roadmap	₹12–18 LPA [futureense]	₹25–35 LPA [futureense]	₹35–50+ LPA [futureense]
With 350+ LC + 7 Projects	₹15–20 LPA	₹30–40 LPA	₹40–60+ LPA

(India Job Market 2026 – Packages & Demand)



QUICK COMPARISON TABLE

Role	Fresher Package	3–5 Years	5+ Years (Senior)	Demand (2026)	Fresher Openings	Learning Time
GenAI Engineer (LLMs, RAG, AI Agents)	₹10–15 LPA	₹20–35 LPA	₹35–60+ LPA	🔥🔥🔥 300% surge	✅ High	6–8 months
Full Stack AI Engineer (GenAI + Full Stack)	₹12–18 LPA	₹18–35 LPA	₹35–60+ LPA	🔥🔥🔥 300% surge	✅✅ Highest	6–8 months
Full Stack AI/ML Engineer (Traditional ML + Full Stack)	₹8–12 LPA	₹15–28 LPA	₹28–45 LPA	🔥🔥 Moderate	⚠️ Medium	12–18 months

Traditional AI/ML Engineer (No Full Stack)	₹6–10 LPA	₹15–25 LPA	₹25–45 LPA	 Moderate	Very Low	12–18 months
Regular Full Stack Engineer (No AI)	₹6–12 LPA	₹12–20 LPA	₹30–40 LPA	 Steady (5% YoY)	  High	4–6 months

The combination of **Software Engineering + AI Integration** is where a lot of hiring demand is moving.

The mistake many students make is trying to become a researcher; most companies actually need engineers who can turn AI capabilities into real products.

Your **MERN** skills are already valuable. The highest-ROI move is usually to add **cloud, deployment, GenAI, and RAG** on top of them rather than pivot completely into traditional AI/ML.

 **YOUR EXACT POSITION (4th Year, Beginner DSA, MERN-only)**

Best Path: Full Stack AI Engineer (GenAI + RAG + Full Stack + Deployment)

Metric	Expectation
Fresher Package	₹12–18 LPA 
5-Year Package	₹35–50 LPA 
Time to Job-Ready	6–8 months 
Difficulty	 Medium (DSA + Backend + GenAI APIs)
Likelihood	 High (most realistic for you)

This is exactly what your 4–5 month roadmap is building. You'll be positioned for **₹12–20 LPA** at product companies, with path to **₹25–35 LPA** in 3–5 years.